Best Times to Travel

Giovanni Cataldi • June 2025

Agenda

- Introduction
- Data Overview
- Visualizations & Key Findings
- Conclusion & Recommendations

When's the Best Time to Visit?

Planning the perfect time to travel means avoiding rainy days, extreme heat, or unexpected cold snaps. This project explores:

- How weather patterns vary across 5 major U.S. cities
- Which months offer the most comfortable conditions
- Where and when travelers should pack their umbrellas or sunscreen

Using historical weather data, I mapped out the ideal travel windows—so you don't have to guess.

Turning Weather Data into Travel Insights

Source

- Historical weather data from Open-Meteo API
- 5 years of hourly records (2020-2025)
- 5 major U.S. cities: New York, Los Angeles, Chicago, Miami, Seattle

What the Data Represents

- **Temperature:** Daily averages, highs, lows
- **Precipitation:** Total rainfall per day
- **Wind Speed:** Average daily wind
- **Humidity:** Comfort factor

Why It Matters

- Scores combine these factors into a **0-100 Comfort Travel Scores**
- Identifies:
- **Best months for ideal weather**
- **Risks** (heatwaves, rainy seasons)
- Enables **city-by-city comparisons**

The Comfort Travel Score

This score is calculated using a mix of different weights of the temperature, precipitation, and wind scores.

- Temperature score, with a weight of 50%, is designed around 68°F, the sweet spot for comfort. 2.22 points are deducted per degree you stray from this ideal.
- Precipitation score, with a weight of 30%—the score loses 10 points per 1 mm of rain, so a day with 5mm of rain cuts 50 points from the max. This harsh penalty reflects how rain disrupts travel plans more than minor temperature changes.
- Wind score, with a weight of 20%—there is a 5 point deduction per 1 m/s, as strong winds make walking uncomfortable but aren't as punishing as rain.

What Does the Data Tell Us?

By analyzing 5 years of weather patterns, the following clear trends were found as to when—and where— travelers can expect the best conditions.

Visualizations & Key Findings

city	Fall	Spring	Summer	Winter	**Total**
Chicago	56.96	51.45	66.87	32.35	**51.91**
Los Angeles	84.57	78.97	85.32	68.29	**79.29**
Miami	57.06	66.45	55.13	71.60	**62.56**
New York	62.47	56.06	68.99	39.75	**56.82**
Seattle	60.29	59.65	80.87	42.45	**60.82**
Total	**64.27**	**62.51**	**71.44**	**50.89**	**62.28**

Figure 1: Average Travel Scores

- Lowest score of 32.35 during Chicago's Winter and highest score of 85.32 during LA's Summer
- LA has consistent high scores (70+)
- Typical for Summer to have the highest score, followed by Fall

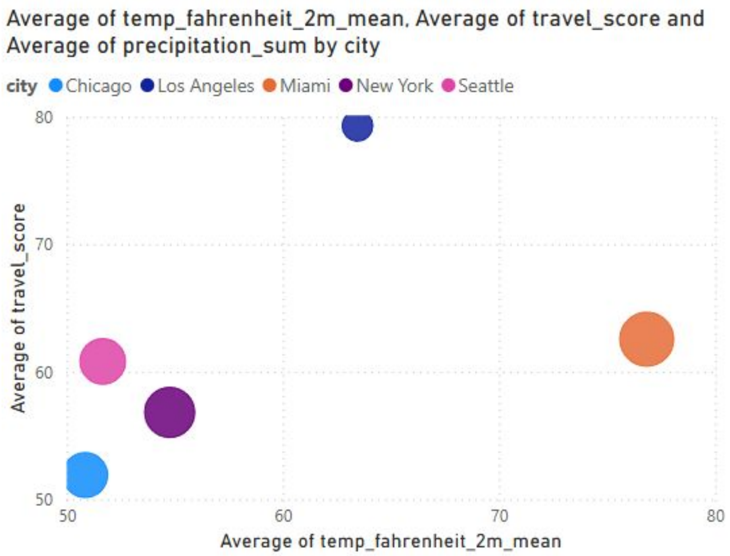

Figure 2: How Temperature and Rainfall Impact Travel Score Across Cities

Larger Circles = Higher Rainfall

- Chicago has the lowest average travel score, with the lowest average temperature and a moderate amount of rainfall
- LA has the highest average travel score, with the an average temperature of ~65 degrees and little rainfall
- While Miami has the highest average temperature, it has the 2nd highest travel score due to large amounts of rainfall
- NYC and Seattle have the 2nd and 3rd lowest average travel score, respectively, due to their lower average temperatures and moderate amounts of rainfall compared to cities scoring higher

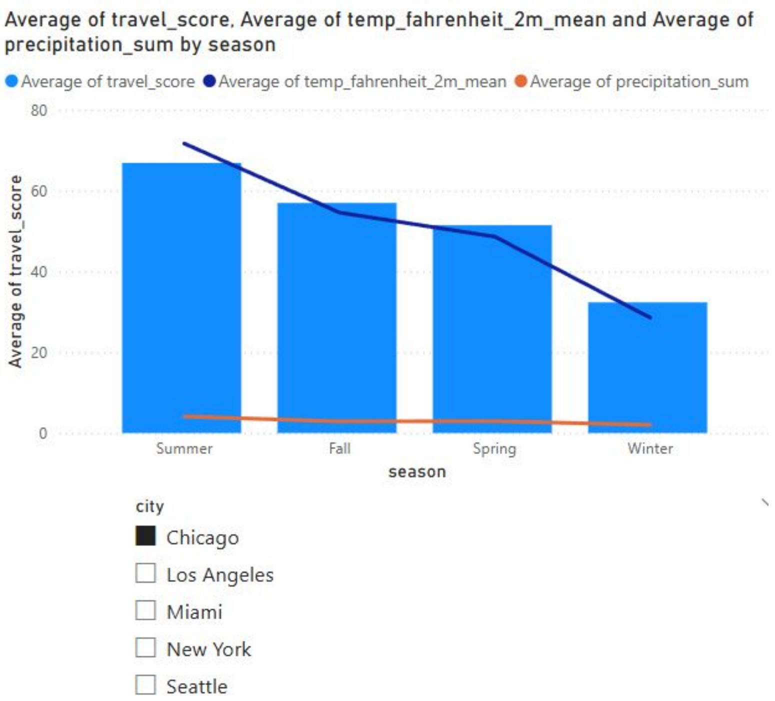

Figure 3: Chicago's Travel Score in Relation to Temperature and Precipitation Amounts

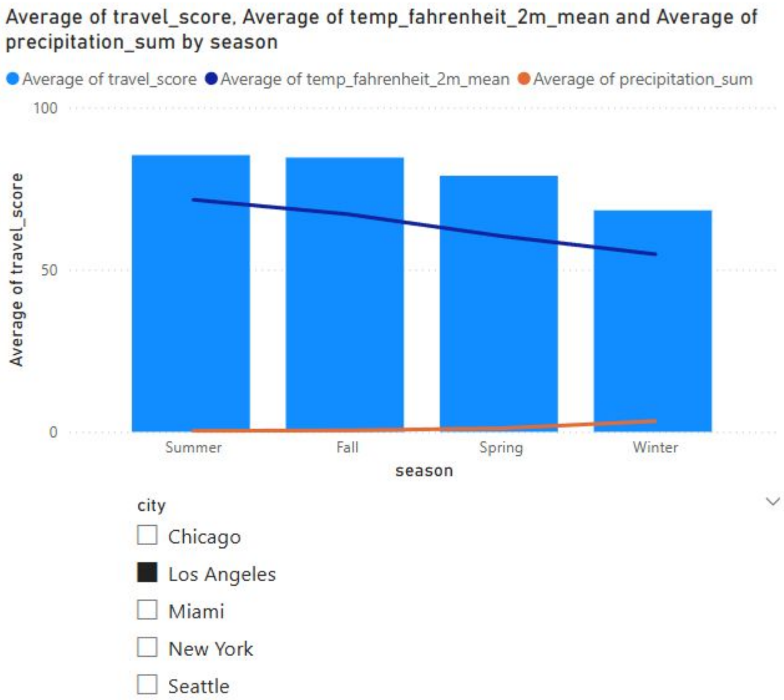

Figure 4: LA's Travel Score in Relation to Temperature and Precipitation Amounts

Chicago and LA

From the previous figures, the contrast in average travel scores between Chicago and Los Angeles is the most striking.

Both cities reach their highest scores in the summer and lowest in winter.

- In Chicago, the average temperature drops gradually from ~70°F in summer to ~30°F in winter, with relatively stable precipitation across seasons.
- In contrast, LA maintains a consistently mild temperature between 60-70°F throughout the year, though it sees a slight increase in rainfall during winter.

These trends suggest that Los Angeles is a strong year-round destination, while Chicago is best visited in the summer for comfortable travel conditions.

Miami

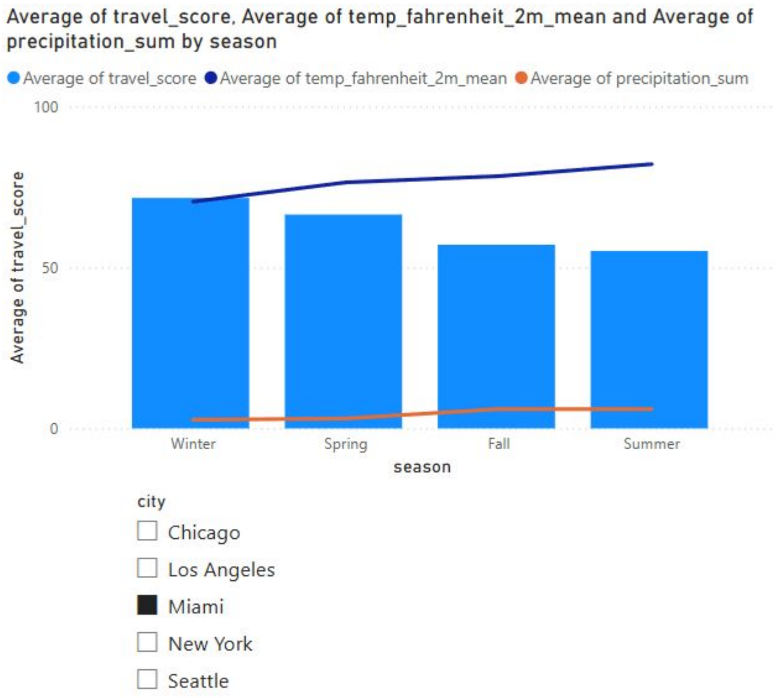

Figure 5: Miami's Travel Score in Relation to Temperature and Precipitation Amounts

Miami presents an interesting exception to the usual seasonal pattern – winter, not summer, appears to be the best time to visit.

- Both temperature and precipitation increase steadily from winter to summer (Winter → Spring → Fall → Summer).
- Summer brings higher heat and more rainfall, leading to increased humidity and a lower travel score.
- Winter, by contrast, offers mild temperatures around 70°F and minimal rainfall, making it the most comfortable season for travel.

These trends suggest that Miami is best enjoyed during the winter and spring months, when conditions are warm but not oppressive.

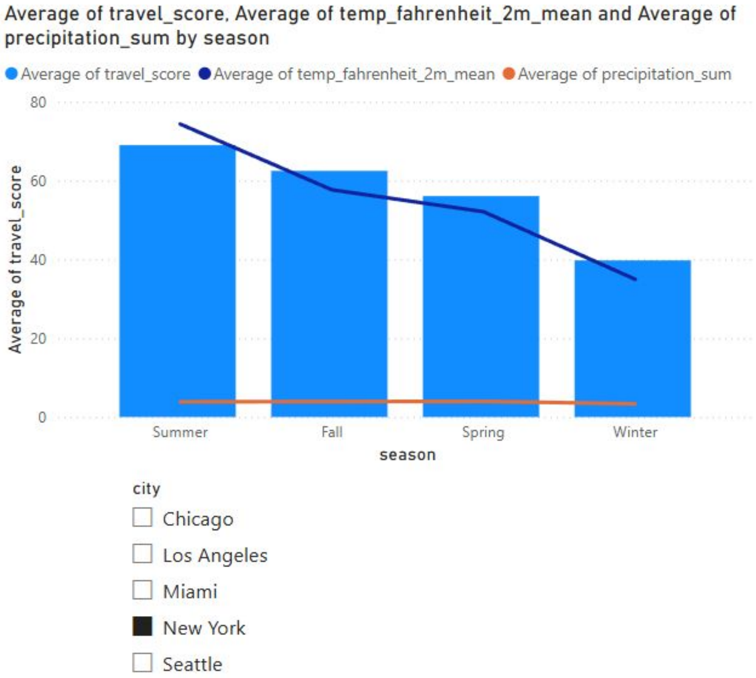

Figure 6: NYC's Travel Score in Relation to Temperature and Precipitation Amounts

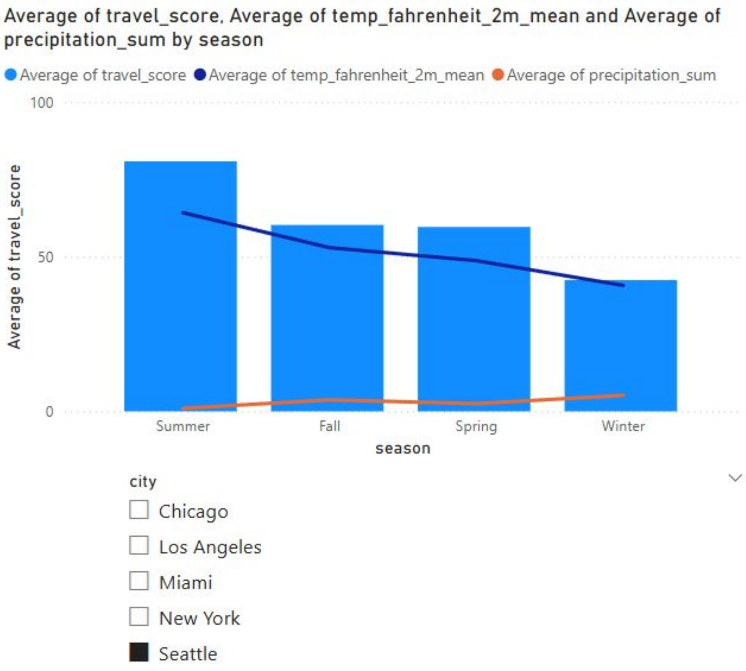

Figure 7: Seattle's Travel Score in Relation to Temperature and Precipitation Amounts

NYC and Seattle

New York City and Seattle follow a similar seasonal pattern, with summer offering the highest average travel scores and Winter the lowest.

- Both cities also see their second-highest score in the fall, suggesting a generally favorable fall.
- NYC experiences a sharper temperature drop across seasons (Summer → Fall → Spring → Winter), while Seattle's temperatures decline more gradually.
- Precipitation in NYC remains relatively stable year-round, whereas Seattle sees increased rainfall in the fall and winter.

Overall, summer is the most favorable season for travel to both NYC and Seattle, with fall as a solid secondary option.

Conclusion & Recommendations

So, what’s the best time to travel?

LA: Any season

Chicago: Summer

Miami: Winter and Spring

NYC: Summer and Fall

Seattle: Summer and Fall